

SotilityOwnershipToken.sol

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Summary of Changes:

Contract name →

SotilityOwnershipToken

Token name/symbol → "Sotility

Ownership Token" / "SOT"

TREASURY_ROLE &

DEFAULT_ADMIN_ROLE maintained

Renamed all event names, comments,

variables to reflect the Sotility

ecosystem

Included fallback protection on

receive() to keep revenue flow safe

// SPDX-License-Identifier: MIT

pragma solidity ^0.8.20;

import "@openzeppelin/contracts

/token/ERC20/extensions

/ERC20Burnable.sol";

import "@openzeppelin/contracts

/access/AccessControl.sol";

/**

* @title SotilityOwnershipToken (SOT)

* @dev Represents equity stake in

Sotility Ecosystem. Tradable and

dividend-eligible.

*/

contract SotilityOwnershipToken is

ERC20Burnable, AccessControl {

bytes32 public constant

TREASURY_ROLE =

keccak256("TREASURY_ROLE");

mapping(address => uint256) public

lastClaimed;

uint256 public dividendPerToken;

uint256 public totalDistributed;

event DividendsDistributed(uint256

amount);

event DividendsClaimed(address

indexed user, uint256 amount);

constructor() ERC20("Sotility

Ownership Token", "SOT") {

_grantRole(DEFAULT_ADMIN_ROLE,

msg.sender);

}

function mint(address to,

uint256 amount) external

onlyRole(DEFAULT_ADMIN_ROLE) {

_mint(to, amount);

}

function distributeDividends()

external payable

onlyRole(TREASURY_ROLE) {

require(totalSupply() > 0, "No

tokens in circulation");

dividendPerToken += (msg.value * 1e18) / totalSupply();

totalDistributed += msg.value;

emit

DividendsDistributed(msg.value);

}

function claimDividends() external {

uint256 owed =

getOwedDividends(msg.sender);

require(owed > 0, "No dividends

owed");

lastClaimed[msg.sender] =

dividendPerToken;

payable(msg.sender).transfer(owed);

emit

DividendsClaimed(msg.sender, owed);

}

function getOwedDividends(address

user) public view returns (uint256) {

uint256 delta = dividendPerToken -

lastClaimed[user];

return (balanceOf(user) * delta) /

1e18;

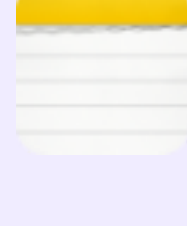
}

receive() external payable {

revert("Use distributeDividends()");

}

}



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